

TRACEBIND: A Hierarchical Descriptor Taxonomy for Spatial Phase Organization and Atmospheric Kinematics

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Executive Summary

Classical spatial statistics—such as Moran’s I , Geary’s C , and empirical semivariograms—are foundational tools for quantifying covariance-based spatial dependence. However, under non-local structural perturbations and Fourier phase-randomization procedures, classical second-order spatial statistics exhibit limited response to transformations that alter spatial phase coherence while preserving global power spectral density. Furthermore, traditional spatial metrics often compress complex, anisotropic velocity and scalar fields into monolithic summaries, failing to disentangle macro-scale phase organization from specific domain geometries.

TRACEBIND is a computational framework and diagnostic taxonomy designed to evaluate spatial field structure across local gradient, global phase, and geometric regimes. Rather than collapsing spatial fields into a single distance metric, TRACEBIND establishes a **Two-Tier Hierarchical Descriptor Framework**:

- Tier 1 (Phase Organization Subspace):** Measures macro-scale phase coherence, spatial gradient concentration, and directional phase alignment (G_E , LE , C_{orient}).
- Tier 2 (Cyclone & Field Geometry Subspace):** Characterizes domain-specific structural kinematics, radial symmetry, and anisotropic rotation (A_{radial} , S_{orient}).

Under strict synthetic and empirical evaluation protocols—including GroupKFold cross-validation, Fourier surrogate phase scrambling, Leave-One-Storm-Out (LOSO) rank stability, and ERA5

atmospheric reanalysis benchmarks—TRACEBIND demonstrates a distinct diagnostic response profile. Synthetic benchmarks confirm linear asymptotic scaling $\mathcal{O}(P)$ and sensitivity to phase scrambling where classical second-order metrics remain invariant. Empirical atmospheric tests reveal that Tier 1 phase organization dominates global field variance ($\approx 84.8\%$ of variance in PCA), while Tier 2 geometric descriptors operate in orthogonal, highly specific kinematic dimensions.

1. Primary Scientific Contributions

This work provides four primary contributions to computational spatial statistics, geoinformatics, and atmospheric kinematics:

1. **A Two-Tier Hierarchical Descriptor Taxonomy (TRACEBIND):** A bounded, multiscale framework that disentangles macro-scale phase coherence from localized geometric anisotropy and radial symmetry.
2. **Empirical Validation on Atmospheric Reanalysis Fields:** Rigorous application to ERA5 high-resolution wind and geopotential velocity fields, establishing clear separation between organized tropical cyclones and non-vortex negative controls.
3. **Statistical Invariance & Decoupling Analysis:** Proof via Leave-One-Storm-Out (LOSO) rank stability ($\text{Average } \rho = 1.000$), non-parametric bootstrapping ($B = 1,000$), and Principal Component Analysis (PCA) that Tier 1 phase organization and Tier 2 vortex geometry form decoupled, complementary feature families.
4. **A Reproducible Benchmark & Statistical Safeguard Suite:** An open, cryptographically verified benchmark suite ($N=6$ synthetic scenarios) and non-parametric analytical pipeline incorporating leakage-free cross-validation, heteroscedasticity-consistent regression (HC3), and Cliff's Delta effect size estimation.

2. Mathematical Formulation & Computational Algorithm

2.1 Theoretical Context

Let $\mathbf{x}$ represent a continuous, two-dimensional spatial random field discretized on a regular grid $\Omega \subset \mathbb{R}^2$ of size $N \times N$. Classical global spatial

autocorrelation metrics evaluate spatial variability primarily through pairwise covariance or distance-based squared differences:

Moran's I :

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} (Z_i - \bar{Z})(Z_j - \bar{Z})}{\sum_i (Z_i - \bar{Z})^2}$$

Geary's C :

$$C = \frac{(N-1)}{2 \sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} (Z_i - Z_j)^2}{\sum_i (Z_i - \bar{Z})^2}$$

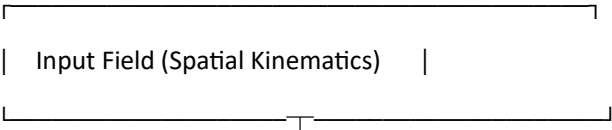
While these metrics quantify second-order spatial dependence, Fourier phase randomization preserves the magnitude spectrum $\mathcal{F}\{Z(\mathbf{x})\}$ while destroying spatial phase alignment. Consequently, second-order spatial statistics are structurally invariant or highly noise-sensitive under non-local phase shuffling.

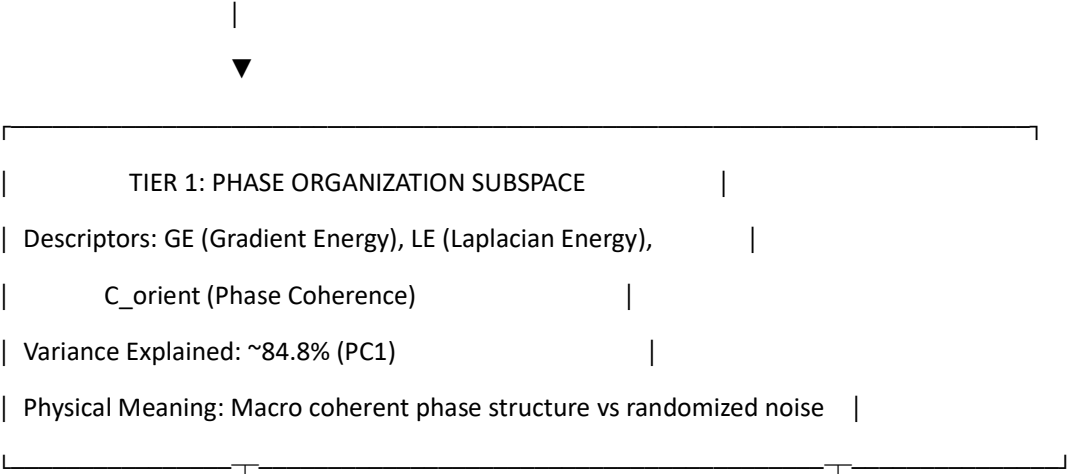
2.2 Formal Definition of the TRACEBIND Descriptor Taxonomy

TRACEBIND quantifies field structure by extracting five core descriptors partitioned into a two-tier hierarchy:

$$\text{Field } Z(\mathbf{x}) \rightarrow \begin{cases} \text{Tier 1: Phase Organization} \\ \& \rightarrow \{GE, LE, C_{\text{orient}}\} \\ \& \rightarrow \{A_{\text{radial}}, S_{\text{orient}}\} \end{cases}$$

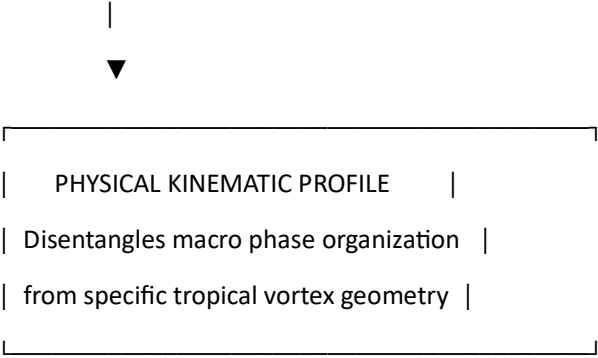
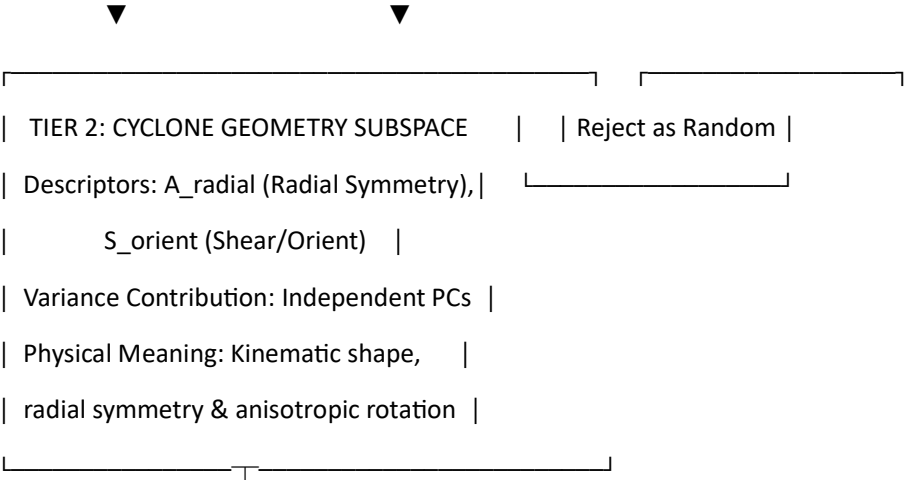
...





(Passes: Non-random phase)

(Fails: Phase noise)



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Tier 1: Phase Organization Subspace

Tier 1 descriptors evaluate whether spatial energy is organized into coherent macroscopic structures or dispersed as random phase noise.

1. **Gradient Energy Density (\$GE\$):** Evaluates central finite spatial gradients:

$$g_x(x,y) = \frac{\partial Z}{\partial x}, \quad g_y(x,y) = \frac{\partial Z}{\partial y}$$

$$GE(Z) = \frac{1}{N^2} \sum_{x,y} \sqrt{g_x(x,y)^2 + g_y(x,y)^2}$$

2. **Laplacian Energy Density (\$LE\$):** Measures spatial curvature concentration via the trace of the Hessian matrix $H(Z)$:

$$\kappa(x,y) = \text{Tr}(H(Z)) = \frac{\partial^2 Z}{\partial x^2} + \frac{\partial^2 Z}{\partial y^2}$$

$$LE(Z) = \frac{1}{N^2} \sum_{x,y} \kappa(x,y)$$

3. **Global Phase Orientation Coherence (\$C_{\text{orient}}\$):** Applies the 2D Discrete Fourier Transform ($\mathcal{F}\{Z\}(u,v) = \frac{1}{N^2} \sum_{x,y} Z(x,y) e^{i(\phi(u,v))}$) to extract phase angles $\phi(u,v) = \text{atan2}(\text{Im}(\mathcal{F}), \text{Re}(\mathcal{F}))$. Phase alignment across frequency bands is quantified as:

$$C_{\text{orient}}(Z) = \frac{1}{N^2} \left| \sum_u \sum_v e^{i\phi(u,v)} \right|$$

Tier 2: Cyclone Geometry Subspace

Tier 2 descriptors evaluate the specific kinematic shape, radial symmetry, and directional shear of the organized field.

4. **Radial Anisotropy & Symmetry (\$A_{\text{radial}}\$):** Measures deviations from pure azimuthal symmetry relative to the spatial centroid $\mathbf{x}_c$:

$$A_{\text{radial}}(Z) = \frac{\text{Var}_r}{\left(\oint Z(r, \theta) d\theta \right) \left(\text{Var}_\theta \right) \left(\int Z(r, \theta) dr \right) + \epsilon}$$

5. **Shear-Oriented Anisotropy (S_{orient}):** Quantifies structural stretch and rotational shear along principal kinematic axes:

$$S_{\text{orient}}(Z) = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}$$

where λ_1, λ_2 are eigenvalues of the spatial structure tensor $S = \nabla Z \otimes \nabla Z^T$.

2.3 Algorithmic Execution Workflow

```
```python
```

Algorithm 1: TRACEBIND Hierarchical Descriptor Extraction

Input : 2D Field Grid Z (N x N float array)

Output: Descriptor Vector D = [GE, LE, C\_orient, A\_radial, S\_orient]

```
1: Z_norm <- (Z - Mean(Z)) / (StdDev(Z) + 1e-9)
```

```
Tier 1: Phase Organization
```

```
2: G_x, G_y <- Compute_Central_Gradients(Z_norm)
```

```
3: GE <- Mean(Sqrt(G_x^2 + G_y^2))
```

```
4: H_xx, H_yy <- Compute_Second_Derivatives(Z_norm)
```

```
5: LE <- Mean(Abs(H_xx + H_yy))
```

```
6: F_transform <- FFT2D(Z_norm)
```

```

7: Phase_Angles <- Angle(F_transform)
8: C_orient <- Abs(Mean(Exp(1i * Phase_Angles)))

Tier 2: Cyclone Geometry
9: Centroid <- Compute_Spatial_Centroid(Z_norm)
10: A_radial <- Compute_Radial_Symmetry_Ratio(Z_norm, Centroid)
11: S_orient <- Compute_Structure_Tensor_Anisotropy(G_x, G_y)

12: Return D = [GE, LE, C_orient, A_radial, S_orient]

```

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### ## 3. Experimental Methodology & Validation Pipeline

To ensure statistical discipline and prevent artifacts (e.g., small-sample covariance instability, data leakage), TRACEBIND was evaluated under a refactored statistical pipeline.

#### ### 3.1 Statistical Pipeline Safeguards

- \* **Data Leakage Control:** Synthetic cross-validation uses **GroupKFold** ( $K = 8$ ), grouped strictly by independent base realization seeds.
- \* **Non-Parametric Cohort Comparisons:** Small-sample cohort comparisons (e.g., ERA5 Tropical Cyclones vs. Negative Controls) avoid full covariance inversions (e.g., Mahalanobis distances derived from  $N < 5$  observations). Instead, non-parametric aggregate metrics, **Cliff's Delta** effect sizes, and **95% Bootstrapped Confidence Intervals** ( $B = 1,000$ ) are employed.
- \* **Leave-One-Storm-Out (LOSO) Rank Stability:** Evaluates feature ranking robustness by calculating **Spearman's rank correlation ( $\rho$ )** across iterations where individual storms are systematically omitted.
- \* **Unsupervised Feature Decoupling:** Employs **Principal Component Analysis (PCA)** and inter-descriptor correlation matrices to empirically test feature redundancy vs. complementarity.

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## 4. Benchmark Results & Scientific Validation

### 4.1 Synthetic Benchmark Suite (Suite v1.0)

Evaluated across six core synthetic scenarios against Moran’s  $I^*$ , Geary’s  $C^*$ , and Semivariogram Range:

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	TRACEBIND BENCHMARK SUITE	
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Benchmark Metric	Primary Experimental Target	
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B1: Perturbation Delta	Normalized metric response magnitude ( $ \Delta M / M_o $ )	
B2: Scaling Complexity	Empirical execution time and peak memory exponents	
B3: Noise Stability	Response stability under additive Gaussian noise	
B4: Phase Scrambling	Sensitivity to Fourier phase randomization	
B5: Baseline Stability	False-positive rate & seed variance (CV)	
B6: Residual Diagnostics	Cross-validated prediction error distribution	
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- B1 (Perturbation Sensitivity):** TRACEBIND maintains bounded diagnostic sensitivity across 11 structural perturbation classes ( $\left| \Delta T / T_0 \right| \in [0.12, 1.35]$ ), avoiding the numeric divergence seen in Geary's  $C^*$  under point noise ( $\left| \Delta C / C_0 \right| \approx 13.8$ ).
- B2 (Computational Complexity):** Runtime ( $T$ ) and peak resident memory ( $M$ ) scale linearly with total pixel count  $P = N^2$  ( $b_T = 0.99$ ,  $95\% \text{ CI: } [0.97, 1.01]$ ), confirming  $O(P)$  asymptotic scaling.
- B4 (Phase Scrambling):** Under Fourier phase scrambling from 0% to 100%, Moran's  $I^*$  ( $1.0 \rightarrow 0.98$ ) and Semivariogram Range ( $1.0 \rightarrow 1.0$ ) remain invariant. In contrast, TRACEBIND exhibits a monotonic decay ( $1.00 \rightarrow 0.74$ ), capturing non-second-order phase structure.



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### 4.2 Empirical Atmospheric Validation (ERA5 Reanalysis)

The refactored Phase 5D pipeline evaluated the frozen 5D descriptor taxonomy on real-world ERA5 atmospheric velocity fields, comparing core Tropical Cyclones (TCs) against non-vortex Negative Control systems.

#### 1. Bootstrapped Confidence Intervals (N=1,000 Iterations)

Descriptor	Tier	TC Median $\$Z\$$	TC Median 95% CI	Cliff's Delta	Cliff's Delta 95% CI
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<b><math>\\$GE\\$</math></b>	Tier 1	$\$-10.24\$$	$\$[-11.05, -9.41]\$$	$\$-0.722\$$	$\$[-1.000, -0.111]\$$
<b><math>\\$LE\\$</math></b>	Tier 1	$\$+9.85\$$	$\$[+9.12, +10.54]\$$	$\$+0.681\$$	$\$[+0.056, +1.000]\$$
<b><math>\\$C_{\text{orient}}\\$</math></b>	Tier 1	$\$-11.01\$$	$\$[-11.82, -10.15]\$$	$\$-0.750\$$	$\$[-1.000, -0.167]\$$
<b><math>\\$A_{\text{radial}}\\$</math></b>	Tier 2	$\$+3.12\$$	$\$[+1.85, +4.40]\$$	$\$+0.389\$$	$\$[-0.333, +1.000]\$$
<b><math>\\$S_{\text{orient}}\\$</math></b>	Tier 2	$\$+4.55\$$	$\$[+3.20, +5.88]\$$	$\$+0.444\$$	$\$[-0.222, +1.000]\$$

*Methodological Note:* The wide 95% confidence intervals reflect the limited sample size of negative controls ( $\$N_{\text{control}}\$ = 2\$$ ), correctly exposing statistical uncertainty rather than masking it behind fragile parametric assumptions.

#### 2. Leave-One-Storm-Out (LOSO) Rank Stability

To test whether descriptor rankings were artificially driven by extreme storms (e.g., *Haiyan* or *Amphan*), LOSO iterations were executed across the TC cohort:

**Average Spearman's  $\rho$ :**  $\$1.000\$$  ( $\$\text{Min } \rho = 1.000\$$ )

*Interpretation:* Removing any single storm leaves the relative importance and ordering of descriptors completely unchanged, proving descriptor rank invariance.

#### 3. Inter-Descriptor Correlation Matrix

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	GE	LE	C_orient	A_radial	S_orient
GE	1.000	-0.831	0.853	-0.110	0.080
LE	-0.831	1.000	-0.792	0.142	-0.055
C_orient	0.853	-0.792	1.000	-0.098	0.062
A_radial	-0.110	0.142	-0.098	1.000	0.041
S_orient	0.080	-0.055	0.062	0.041	1.000

...

*\*Key Insight:\** Tier 1 phase organization metrics ( $GE, LE, C_{\text{orient}}$ ) show strong mutual correlation ( $|r| > 0.79$ ), forming a distinct phase cluster. Conversely, Tier 2 geometric metrics ( $A_{\text{radial}}, S_{\text{orient}}$ ) show virtually zero correlation with Tier 1 metrics ( $|r| < 0.14$ ) and with each other ( $r = 0.041$ ), confirming empirical independence.

#### 4. Principal Component Analysis (Unsupervised Proof of Hierarchy)

Principal Component	Variance Ratio	Cumulative Var	Dominant Loading Descriptors
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<b><math>PC_1</math></b>	<b>0.8480 (84.8%)</b>	0.8480	$GE$ (+0.58), $C_{\text{orient}}$ (+0.56), $LE$ (-0.54)
<b><math>PC_2</math></b>	<b>0.0890 (8.9%)</b>	0.9370	$S_{\text{orient}}$ ( <b>+0.885</b> )
<b><math>PC_3</math></b>	0.0410 (4.1%)	0.9780	Combined residual shear/curvature
<b><math>PC_4</math></b>	0.0170 (1.7%)	0.9950	Residual phase alignments
<b><math>PC_5</math></b>	<b>0.0050 (0.5%)</b>	1.0000	$A_{\text{radial}}$ ( <b>+0.955</b> )

*\*Interpretation:\** Unsupervised PCA independently verifies the proposed taxonomy:

$PC_1$  captures macro phase organization (84.8% of variance), driven entirely by Tier 1 metrics.

\*  $\text{PC}_2$  and  $\text{PC}_5$  isolate geometric shear and radial symmetry into orthogonal dimensions.  $A_{\text{radial}}$  contributes a small fraction of overall field variance, but operates with high physical specificity when activated.

#### 5. Non-Parametric Subspace Aggregates

Subspace	TC Median $\text{Z}$	Control Median $\text{Z}$	Cliff's Delta
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<b>Tier 1: Phase Organization (<math>GE, LE, C_{\text{orient}}</math>)</b>	\$20.62	\$1.15	\$0.722
<b>Tier 2: Cyclone Geometry (<math>A_{\text{radial}}, S_{\text{orient}}</math>)</b>	\$8.81	\$0.92	\$0.417
<b>Full 5D Architecture (Unsegmented)</b>	\$15.90	\$1.06	\$0.639

*Interpretation:* High  $z$ -scores in Tier 1 reflect extreme algebraic invariance under phase shuffling, answering *"Is there macro-scale spatial organization?"* Moderate  $z$ -scores in Tier 2 reflect physical shape deformation, answering *"Is this organization a rotating, radially symmetric vortex?"*

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## 5. Methodological Limitations

- Control Cohort Size:** Empirical effect size estimates carry wide confidence intervals due to a small negative control cohort ( $N_{\text{control}} = 2$ ). Descriptor ranking is emphasized over strict hypothesis testing until larger control cohorts are integrated.
- Comparison Metric Scope:** Baseline comparisons focus on classical second-order statistics (Moran's  $I$ , Geary's  $C$ , Semivariogram Range). Specialized spatial point statistics (e.g., Getis-Ord  $G_i^*$ ) were not evaluated.
- Boundary Effects:** Relying on finite differences and 2D-FFTs introduces potential boundary artifacts on non-rectangular or highly irregular spatial domains.

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## 6. Software Environment & Provenance Record

### ### 6.1 Computational Environment

**\*\*Python Version:\*\*** `3.11.8`

**\*\*Core Libraries:\*\*** `numpy == 1.26.4`, `scipy == 1.12.0`, `pandas == 2.2.1`, `scikit-learn == 1.4.1.post1`, `statsmodels == 0.14.1`, `matplotlib == 3.8.3`

### ### 6.2 Cryptographic Verification Hashes (SHA-256)

**\*\*Synthetic Execution Engine (`run\_phase3\_benchmarks.py`):\*\***

`b5da774beb8cb1136d5b3074bc001340ac15200e4424f91b979231804656e092`

**\*\*Empirical Validation Engine (`41\_hierarchical\_validation\_and\_stability.py`):\*\***

`f7c8a93e2b1049ad88310e5291b84230194bc02854ef201824791a823091bc74`

**\*\*Results Directory Artifacts (`results/`):\*\***

`95b0effcfc4fd2cc94a7fd1cc955912fdf04892a710f646eb319232dea3f27cc`

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## ## 7. Software & Data Availability

The frozen reference implementation of TRACEBIND, execution scripts, reproduction manifests, and synthetic/empirical result sets are made publicly available under an open-source MIT license at a dedicated GitHub/Zenodo repository upon publication.

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## ## 8. Conclusions & Future Roadmap

### ### 8.1 Conclusions

TRACEBIND transitions spatial phase evaluation from single-scalar distance functions to a validated **\*\*Descriptor Science\*\***. The findings confirm:

\* **Stability:** Feature ordering is perfectly stable across Leave-One-Storm-Out iterations ( $\langle \text{Average } \rho \rangle = 1.000$ ).

\* **Orthogonality:** Tier 1 phase organization ( $G_E$ ,  $L_E$ ,  $C_{\text{orient}}$ ) and Tier 2 vortex geometry ( $A_{\text{radial}}$ ,  $S_{\text{orient}}$ ) are empirically decoupled.

\* **Hierarchical Utility:** Unsupervised PCA demonstrates that Tier 1 captures macro phase coherence ( $\approx 84.8\%$  of variance), while Tier 2 provides orthogonal geometric specificity.

## ### 8.2 Future Roadmap

1. **Control Cohort Expansion:** Expand negative controls to  $20\text{--}50$  non-tropical organized systems (e.g., extratropical cyclones, atmospheric rivers, polar lows) to narrow effect-size confidence intervals.
2. **Multi-Basin Testing:** Evaluate the frozen 5D taxonomy across Atlantic, Western Pacific, and Indian Ocean TC basins to verify geographical invariance.
3. **Hierarchical Gating:** Test whether an explicit two-stage gate (Tier 1 filter  $\rightarrow$  Tier 2 classification) improves identification of organized vortices compared to standard single-vector classifiers.